

Kryptos K4 Analysis

Apologies if any of the following have been discussed before and I am simply rediscovering old insights, but I thought some of these were interesting and hopefully might contain some ideas and observations worth exploring.

I am a big fan of the 21 column layout that many have suggested, and have been studying it for a while in various forms. Most of the observations in this document are centered around that format as well as alphabet streams in some form or another.

A small introduction to both the format and the alphabet streams, is the “**JKL**” that appears in the dead center of the chart (both vertically and horizontally in the 21 column layout. The **K** is repeated both above and below the center **K** in the same column, and the JLK lines up perfectly with the standard alphabet starting from column 1. By itself this is just sort of an interesting pattern, but will become more interesting later looking more into alphabet streams and their relationship to the ciphertext (and the plaintext!)

Images and descriptions below:

Index	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Column	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21
Plaintext																					
Ciphertext	O	B	K	R	U	O	X	O	G	H	U	L	B	S	O	L	I	F	B	B	W

Index	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41
Column	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21
Plaintext	E	A	S	T	N	O	R	T	H	E	A	S	T								
Ciphertext	F	L	R	V	Q	Q	P	R	N	G	K	S	S	O	T	W	T	Q	S	J	Q

A B C D E F G H I J (K) L

Index	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62
Column	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21
Plaintext																					
Ciphertext	S	S	E	K	Z	Z	W	A	T	J	K	L	U	D	I	A	W	I	N	F	B

DEAD CENTER

Index	63	64	65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83
Column	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21
Plaintext	B	E	R	L	I	N	C	L	O	C	K										
Ciphertext	N	Y	P	V	T	T	M	Z	F	P	K	W	G	D	K	Z	X	T	J	C	D

Index	84	85	86	87	88	89	90	91	92	93	94	95	96								
Column	1	2	3	4	5	6	7	8	9	10	11	12	13								
Plaintext																					
Ciphertext	I	G	K	U	H	U	A	U	E	K	C	A	R								

In this post, the author describes an interesting phenomenon where duplicate plaintext letters appear in the ciphertext in line with the alphabet. For example LINCL -> VTTMZ

<https://pcsxctrasupport3.wordpress.com/2016/08/04/my-first-deep-look-at-kryptos-k4/>

I decided to experiment with different alphabet streams to see how they lined up with the ciphertext. I believe the above post was created before the (EAST)NORTHEAST cribs were disclosed. Interestingly the author's observation not only continues into that section of the cipher, but if you overlay an oscillating alphabet (A to Z to A to Z, etc...) with the text, it lines up with both cribs perfectly in columns 4 and 8. (An almost similar pattern forms in both sections in columns 12 and 16 as well)

Index	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41
Column	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21
Plaintext	E	A	S	T	N	O	R	T	H	E	A	S	T								
Alphabet 1	Y	X	W	V	U	T	S	R	Q	P	O	N	M	L	K	J	I	H	G	F	E
Ciphertext	F	L	R	V	Q	Q	P	R	N	G	K	S	S	O	T	W	T	Q	S	J	Q

S T U V W

Index	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62
Column	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21
Plaintext																					
Alphabet 1	D	C	B	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R
Ciphertext	S	S	E	K	Z	Z	W	A	T	J	K	L	U	D	I	A	W	I	N	F	B

Index	63	64	65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83
Column	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21
Plaintext	B	E	R	L	I	N	C	L	O	C	K										
Alphabet 1	S	T	U	V	W	X	Y	Z	Y	X	W	V	U	T	S	R	Q	P	O	N	M
Ciphertext	N	Y	P	V	T	T	M	Z	F	P	K	W	G	D	K	Z	X	T	J	C	D

W X Y Z (ALMOST)

Additionally, if you overlay 2 opposing alphabets, you can see that some of the ciphertext lines up with the reverse alphabet as well:

Index	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41
Column	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21
Plaintext	E	A	S	T	N	O	R	T	H	E	A	S	T								
Alphabet 1	Y	X	W	V	U	T	S	R	Q	P	O	N	M	L	K	J	I	H	G	F	E
Alphabet 2	V	U	T	S	R	Q	P	O	N	M	L	K	J	I	H	G	F	E	D	C	B
Ciphertext	F	L	R	V	Q	Q	P	R	N	G	K	S	S	O	T	W	T	Q	S	J	Q

This would not be so remarkable by itself, but it continues into the BERLINCLOCK crib as well.

This next chart gets a little messy, but in it you can see that:

- In columns 4 and 8, we have TNORT lining up with VUTSR
- In columns 4 and 8, we have LINCL lining up with VWXYZ
- In columns 7 through 10, we have the ciphertext PRNG lining up with PONM
- In columns 7 through 10, we have the ciphertext MZFP lining up with MNOP

It may all be coincidence, but it does seem to reveal a consistent pattern that begins to look pretty non-random. It begins to look even less random when you look into the positions where this is occurring relative to the overall index. (The abscissa?). The mid-point of many of these sections seem to land at multiples of 13 or 17, both prime.

If not directly tied to the structure of algorithm itself, then possibly a window into the masking technique.

Index	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Column	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21
Plaintext																					
Alphabet 1	H	G	F	E	D	C	B	A	B	C	D	E	F	G	H	I	J	K	L	M	N
Alphabet 2	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
Alphabet 3	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	Y	X	W
Ciphertext	O	B	K	R	U	O	X	O	G	H	U	L	B	S	O	L	I	F	B	B	W
Index	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41
Column	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21
Plaintext	E	A	S	T	N	O	R	T	H	E	A	S	T								
Alphabet 1	O	P	Q	R	S	T	U	V	W	X	Y	Z	Y	X	W	V	U	T	S	R	Q
Alphabet 2	Y	X	W	V	U	T	S	R	Q	P	O	N	M	L	K	J	I	H	G	F	E
Alphabet 3	V	U	T	S	R	Q	P	O	N	M	L	K	J	I	H	G	F	E	D	C	B
Ciphertext	F	L	R	V	Q	Q	P	R	N	G	K	S	S	O	T	W	T	Q	S	J	Q
Index	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62
Column	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21
Plaintext																					
Alphabet 1	P	O	N	M	L	K	J	I	H	G	F	E	D	C	B	A	B	C	D	E	F
Alphabet 2	D	C	B	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R
Alphabet 3	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U
Ciphertext	S	S	E	K	Z	Z	W	A	T	J	K	L	U	D	I	A	W	I	N	F	B
Index	63	64	65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83
Column	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21
Plaintext	B	E	R	L	I	N	C	L	O	C	K										
Alphabet 1	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	Y
Alphabet 2	S	T	U	V	W	X	Y	Z	Y	X	W	V	U	T	S	R	Q	P	O	N	M
Alphabet 3	V	W	X	Y	Z	Y	X	W	V	U	T	S	R	Q	P	O	N	M	L	K	J
Ciphertext	N	Y	P	V	T	T	M	Z	F	P	K	W	G	D	K	Z	X	T	J	C	D
Index	84	85	86	87	88	89	90	91	92	93	94	95	96								
Column	1	2	3	4	5	6	7	8	9	10	11	12	13								
Plaintext																					
Alphabet 1	X	W	V	U	T	S	R	Q	P	O	N	M	L	H	G	F	E	D	C	B	A
Alphabet 2	L	K	J	I	H	G	F	E	D	C	B	A	B	F	G	H	I	J	K	L	M
Alphabet 3	I	H	G	F	E	D	C	B	A	B	C	D	E	I	J	K	L	M	N	O	P
Ciphertext	I	G	K	U	H	U	A	U	E	K	C	A	R								

13

Index	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Column	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21
Plaintext												?				?					
Ciphertext	O	B	K	R	U	O	X	O	G	H	U	L	B	S	O	L	I	F	B	B	W

2x13 2x17

Index	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41
Column	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21
Plaintext	E	A	S	T	N	O	R	T	H	E	A	S	T			?					
Ciphertext	F	L	R	V	Q	Q	P	R	N	G	K	S	S	O	T	W	T	Q	S	J	Q

V U T S A S T U V W

Index	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62
Column	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21
Plaintext																					
Ciphertext	S	S	E	K	Z	Z	W	A	T	J	K	L	U	D	I	A	W	I	N	F	B

4x17

Index	63	64	65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83
Column	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21
Plaintext	B	E	R	L	I	N	C	L	O	C	K										
Ciphertext	N	Y	P	V	T	T	M	Z	F	P	K	W	G	D	K	Z	X	T	J	C	D

V W X Y Z

Index	84	85	86	87	88	89	90	91	92	93	94	95	96								
Column	1	2	3	4	5	6	7	8	9	10	11	12	13								
Plaintext																					
Ciphertext	I	G	K	U	H	U	A	U	E	K	C	A	R								

Some other observations that may or may not be noise. If we assume that the word THE comes before BERLINCLOCK, then there is an interesting relationship to the oscillating alphabet and the THE in EASTNOTHEAST:

Index	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41
Column	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21
Plaintext	E	A	S	T	N	O	R	T	H	E	A	S	T								
Alphabet 1	Y	X	W	V	U	T	S	R	Q	P	O	N	M	L	K	J	I	H	G	F	E
Ciphertext	F	L	R	V	Q	Q	P	R	N	G	K	S	S	O	T	W	T	Q	S	J	Q

Index	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62
Column	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21
Plaintext																					
Alphabet 1	D	C	B	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R
Ciphertext	S	S	E	K	Z	Z	W	A	T	J	K	L	U	D	I	A	W	I	N	F	B

Index	63	64	65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83
Column	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21
Plaintext	B	E	R	L	I	N	C	L	O	C	K										
Alphabet 1	S	T	U	V	W	X	Y	Z	Y	X	W	V	U	T	S	R	Q	P	O	N	M
Ciphertext	N	Y	P	V	T	T	M	Z	F	P	K	W	G	D	K	Z	X	T	J	C	D

There is some interesting symmetry(/potential stego?) that may be nothing but it might be worth looking into as the pattern does tie into the middle sections of the other patterns in places (The Q in column 26, and the T in column 6):

Index	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Column	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21
Plaintext																					
Ciphertext	O	B	K	R	U	O	X	O	G	H	U	L	B	S	O	L	I	F	B	B	W

Index	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41
Column	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21
Plaintext	E	A	S	T	N	O	R	T	H	E	A	S	T								
Ciphertext	F	L	R	V	Q	Q	P	R	N	G	K	S	S	O	T	W	T	Q	S	J	Q

Index	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62
Column	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21
Plaintext																					
Ciphertext	S	S	E	K	Z	Z	W	A	T	J	K	L	U	D	I	A	W	I	N	F	B

Index	63	64	65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83
Column	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21
Plaintext	B	E	R	L	I	N	C	L	O	C	K										
Ciphertext	N	Y	P	V	T	T	M	Z	F	P	K	W	G	D	K	Z	X	T	J	C	D

Index	84	85	86	87	88	89	90	91	92	93	94	95	96								
Column	1	2	3	4	5	6	7	8	9	10	11	12	13								
Plaintext																					
Ciphertext	I	G	K	U	H	U	A	U	E	K	C	A	R								

I discovered this by accident because I was using the wrong indices in the KRYPTOS tableau, but for certain plaintext letters, there is a relationship to the ciphertext letters using the oscillating alphabet and the A-Z indices on the tableau:

Index	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Column	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21
Plaintext																					
Alphabet 1	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
Ciphertext	O	B	K	R	U	O	X	O	G	H	U	L	B	S	O	L	I	F	B	B	W

Index	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41
Column	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21
Plaintext	E	A	S	T	N	O	R	T	H	E	A	S	T								
Alphabet 1	Y	X	W	V	U	T	S	R	Q	P	O	N	M	L	K	J	I	H	G	F	E
Ciphertext	F	L	R	V	Q	Q	P	R	N	G	K	S	S	O	T	W	T	Q	S	J	Q

USING STANDARD ALPHABET INDICES ON THE KRYPTOS TABLEAU
 E, X = R A, W = V T, L = T T, N = E

Index	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62
Column	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21
Plaintext																			T	H	E
Alphabet 1	D	C	B	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R
Ciphertext	S	S	E	K	Z	Z	W	A	T	J	K	L	U	D	I	A	W	I	N	F	B

There seems to be a relationship between Columns 1 and 10 in the form of:

(ciphertext = plaintext + offset + |column / 10|)

Index	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41
Column	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21
Plaintext	E	A	S	T	N	O	R	T	H	E	A	S	T								
Ciphertext	F	L	R	V	Q	Q	P	R	N	G	K	S	S	O	T	W	T	Q	S	J	Q

$$C = P + 1$$

$$C = P + 2$$

Index	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62
Column	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21
Plaintext																					
Ciphertext	S	S	E	K	Z	Z	W	A	T	J	K	L	U	D	I	A	W	I	N	F	B

Index	63	64	65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83
Column	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21
Plaintext	B	E	R	L	I	N	C	L	O	C	K										
Ciphertext	N	Y	P	V	T	T	M	Z	F	P	K	W	G	D	K	Z	X	T	J	C	D

$$C = P + 12$$

$$C = P + 13$$

...and last, and probably least, A few of the ciphertext doublings are followed by the next letter in the alphabet (mod 26) 1 character away. If you want to get really “tin foil hat”, you can probably connect P, Q, R, S, and T somehow LOL.

Index	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41
Column	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21
Plaintext	E	A	S	T	N	O	R	T	H	E	A	S	T								
Ciphertext	F	L	R	V	Q	Q	P	R	N	G	K	S	S	O	T	W	T	Q	S	J	Q

Index	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62
Column	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21
Plaintext																					
Ciphertext	S	S	E	K	Z	Z	W	A	T	J	K	L	U	D	I	A	W	I	N	F	B

A couple of side notes that I forgot to mention above as well:

- There does seem to be some portions of the ciphertext that line up with repeated KRYPTOS forward and backward, as well as the KRYPTOS alphabet forward and backward.
- On page 2 of [Military Crypanalysis III](#) there is a “Berlin clock” looking cipher described.
- Also the logo of Ed Scheidt’s company “Tec Sec” has a Berlin clock theme to it (as well as his driveway which I’m sure you’re all aware):

